ENCE260: Embedded Systems

Lecture 5: Multiplexing

When you just don't have enough pins.

Revision

Will this code work for all prescalar values?

```
#define F_CLK 8000000
#define TIMER1_PRESCALE 1024

uint16_t delay = 500; // Milliseconds.

uint16_t ticks = delay*(F_CLK/TIMER1_PRESCALE)/1000;

// Run delay using ticks...
```

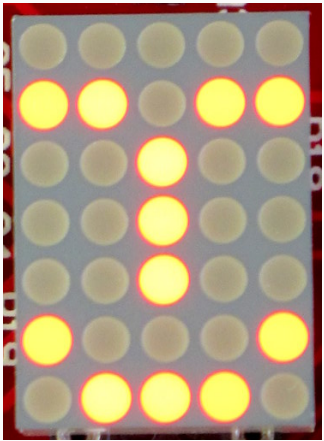
1024: (8000000/1024) = 7812, 500*7812 = 3906500, 3906500/1000 = 3906

⋮

64: (8000000/64) = 125000, 500*125000 = 62500000, 15625000/1000 = 15625

8: (8000000/8) = 1000000, 500*1000000 = 500000000, 500000000/1000 = 500000, > 65535

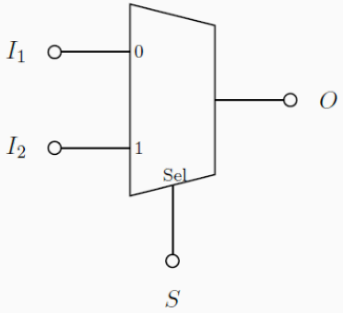
LED Matrix



- Instead of an actual “screen”, the Fun Kit has a matrix of LEDs.
- The LED matrix has 7 rows and 5 columns.
- This is a total of 35 individual LEDs.
- The ATmega32U2 only has 23 PIO pins, though some micros have more.
- For real screens (e.g. $1920 \times 1080 = 2073600$ pixels), direct drive simply isn’t an option.
- PIO pins are expensive! We need some other way.

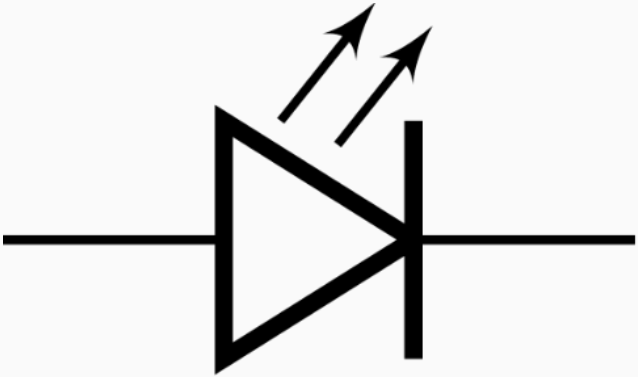
Multiplexers

- A *multiplexer* is a switching circuit that provides a path from one of a selection of inputs to the output.
- These can be constructed from AND and OR gates.
- With N control bits, we can choose between 2^N inputs.

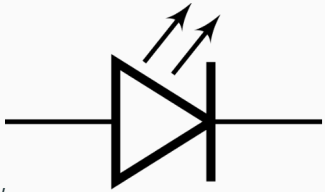


Light Emitting Diodes (LEDs)

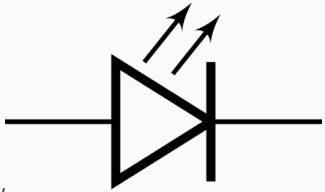
- An LED is on when current flows through it.
- The main arrow indicates the direction of current flow.



Turning on an LED



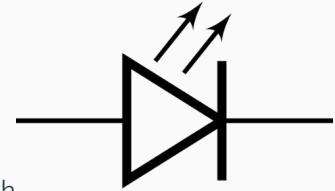
Low Low



Low High



High Low



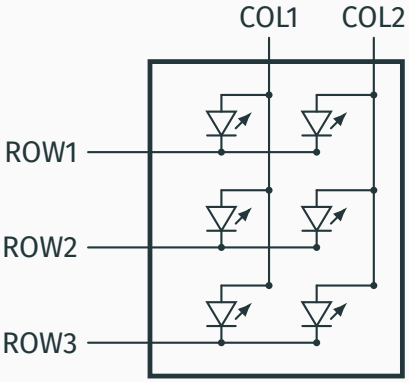
High High

LED Matrix Construction

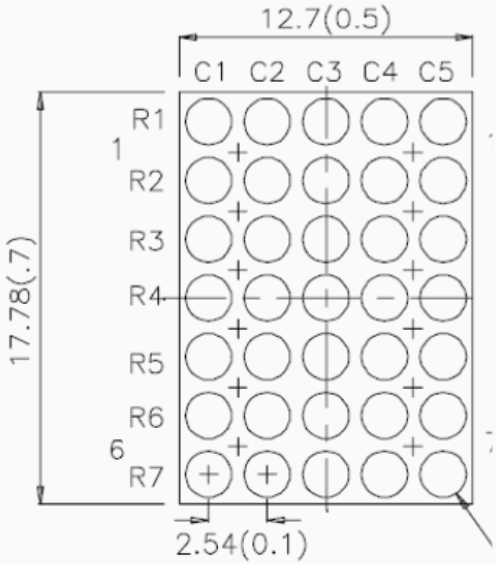
- The LED matrix instead connects the anodes in columns, and cathodes in rows.
- Now we need only (rows + columns) PIO pins, not (rows $\times$ columns) PIO pins.
- For the 7×5 Fun Kit display:

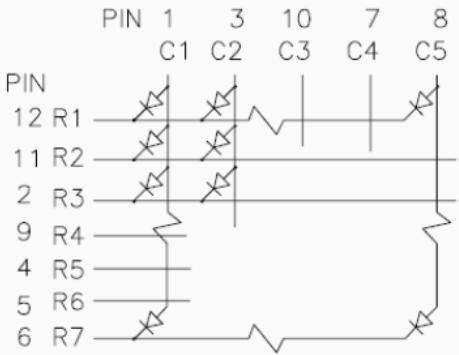
  Old: $7 \times 5 = 35$ pins.

  New: $7 + 5 = 12$ pins.
- Typically, to turn on an individual LED, we set the corresponding column high and row low.



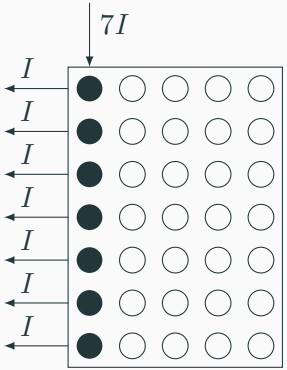
The Fun Kit's LED Matrix





Current Considerations

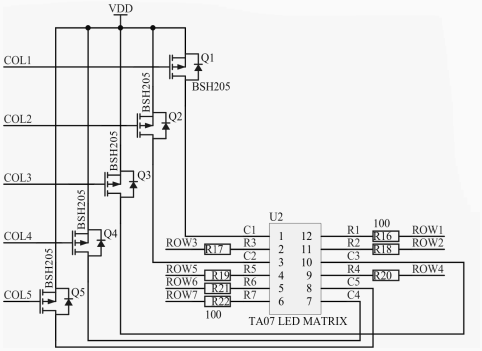
- Consider what happens when we turn on all the LEDs in one column.



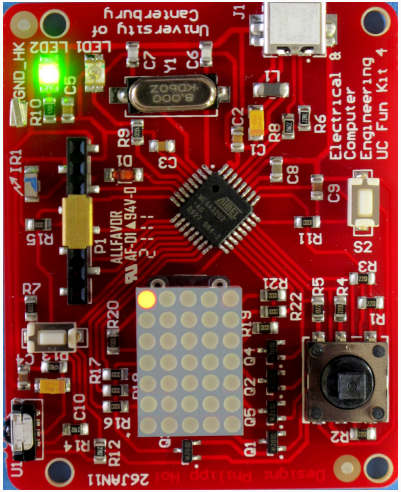
- This is more current than a single PIO pin on the ATmega32U2 can provide.

MOSFETs to the Rescue

- We use p-channel Metal Oxide Semiconductor Field Effect Transistors (MOSFETs) to drive the columns.
- These FET drivers are turned on by inverse logic: on a low input, they connect to the $5\,\mathrm{V}$ supply, otherwise they become open circuit.
- This has the side benefit of LEDs turning on when both pins are low (instead of being different).



Driving the UCFK's LED Matrix




```c
#include "pio.h"
#include "system.h"

int main(void)
{
    system_init();

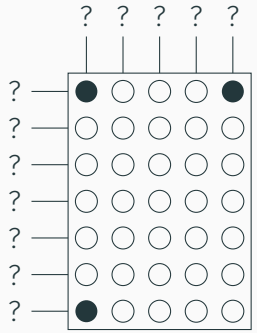
    // Configure the PIOs.
    pio_config_set(LEDMAT_ROW1_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_ROW2_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_ROW3_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_ROW4_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_ROW5_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_ROW6_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_ROW7_PIO, PIO_OUTPUT_HIGH);

    pio_config_set(LEDMAT_COL1_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_COL2_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_COL3_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_COL4_PIO, PIO_OUTPUT_HIGH);
    pio_config_set(LEDMAT_COL5_PIO, PIO_OUTPUT_HIGH);

    while (1) {
        // Turn on the top-left LED.
        pio_output_low(LEDMAT_ROW1_PIO);
        pio_output_low(LEDMAT_COL1_PIO);
    }
}
```

The Three Corner Problem

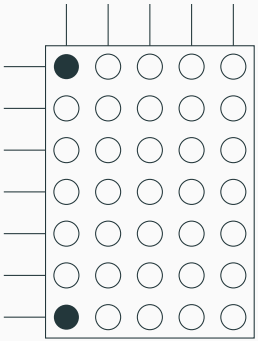
- How do we illuminate the LEDs in three corners of the matrix, but not the fourth?

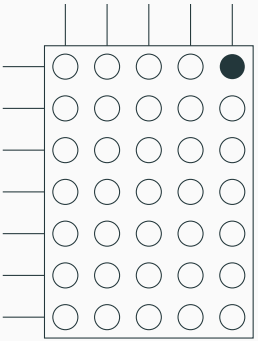


- We need to time division multiplex the columns to trick the eye into thinking the LEDs are on and off.

The Three Corner Problem Solution

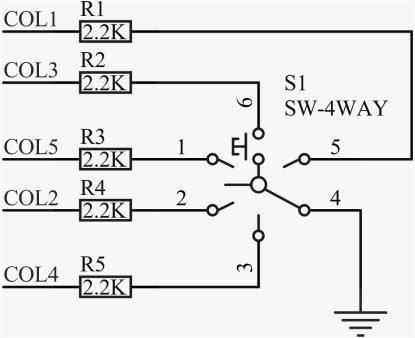
- We toggle between two states, spending half of our time in each state.
- This is called “Time-division multiplexing”.
- On the UCFK4, LED array element is on when its column & row pins are both low.





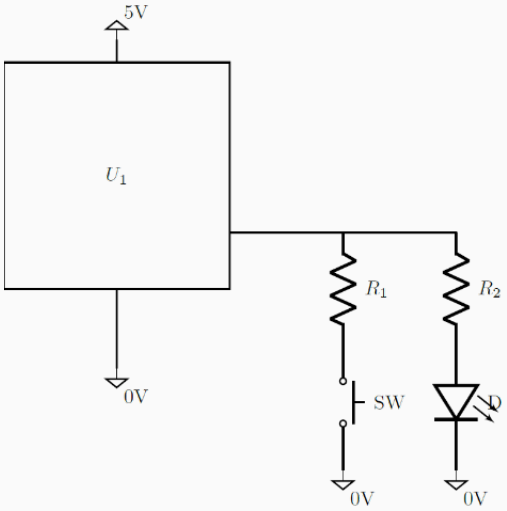
Navigational Switch

- Even after multiplexing output pins to the display, we still don't have enough pins for all the inputs.
- The navigation switch is five switches (north, east, south, west, down) in the same package with a common connection.



Sharing Pin Functionality

- The navigation switch (inputs) are shared with the LED matrix columns (outputs).
- Most of the time, the PIO pin is running as an output driving the LED.
- To read the push button, we switch the pin to an input (and enable the internal pull-up resistor)



Key Concepts

- We often don't have enough pins to directly control everything we might want to.
- We can use dedicated hardware to solve this problem, but if possible it's better to share pins by time-division multiplexing.
- On the Fun Kit, we use persistence of vision to drive the LED matrix with many fewer pins.
- We can also time-division multiplex pins between data directions to use them as both inputs and outputs.

Homework

What are the side-effects of the multiplexing used on the UCFK?

Both the display column multiplexing and data direction multiplexing have their own effects, so try to think of one for each.